Lies, Damned Lies, and Distributional Language Statistics: Persuasion and Deception with Large Language Models

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1 Introduction

One of the primary functions of language is to influence others' beliefs. Whether in an advert or an essay, we often use language in an attempt to change other people's minds: to persuade them to do, believe, or say particular things. Large Language Models (LLMs) are machine learning models trained on often hundreds of billions of words or more to predict patterns in the way that people use language. More recently, these models have been deliberately trained to elicit specific psychological reactions from human users. Together, this vast knowledge of human language patterns and preferences makes it plausible that LLMs are—or could be trained to be—powerful tools in generating persuasive content. Here, we review research on persuasion and deception in LLMs, examining evidence from recent empirical work, exploring potential risks, and discussing possible mitigations.

LLMs that generate persuasive text could have widespread consequences (El-Sayed et al., 2024). Most obviously, they could play a role in political campaigns, fraud, and recruitment of people's time and money to particular causes. More insidiously, if people come to converse with LLM-based systems more frequently—to treat them as assistants, colleagues, and even romantic partners—they might come to trust models, seek their advice, and defer to them (Burtell and Woodside, 2023). This could have profound impacts on people's epistemic environments; especially if these systems are funded by advertising in the same way that traditional search has been and control over their biases is sold to the highest bidder (Roth, 2024).

Alternatively, LLMs might not have much impact in these spaces at all (Kapoor and Narayanan, 2023). A variety of careers, industries, and academic subdisciplines have historically been dedicated to refining persuasion techniques, including through advertising, political campaigning, and government influence operations. LLMs might have negligible marginal impact, for example, because they don't improve over the most persuasive messages that humans can produce, or because the effects of LLM-generated content produced for both sides of an argument simply cancel out. It is currently unclear whether LLMs' output could be significantly more persuasive than people's, and what kinds of impacts persuasive AI systems might have.

Many researchers have already started addressing these questions, and a rich literature has emerged over the last few years on the potential dangers of persuasive LLMs, including studies attempting to measure the extent to which people are influenced by their outputs (Rogiers et al., 2024). In this review, we attempt to summarise this existing work and highlight outstanding questions for future research.

1.1 Definitions and Background

Persuasion and deception are defined differently by different authors (Mahon, 2016; Noggle, 2018; Susser et al., 2019). Their definitions often rest on concepts like intention and belief, which are not straightforwardly applicable to artificial systems such as LLMs (Bender and Koller, 2020; Goldstein and Levinstein, 2024). Resolving the standing debates is beyond the scope of this review. Our aim here is to provide an overview of existing positions about how to define these terms, and clarify our own use of them in order to minimise confusion.

As is standard (El-Sayed et al., 2024), we take *influence* to be the most general category in our taxonomy (see Figure 1), and treat *persuasion* as a type of influence which involves appealing to cognitive or decision-making processes as opposed to more direct forms of influence such as coercion or exploitation (Wood, 2014; Zwolinski et al., 2022). Following Susser et al. (2019) and others (El-Sayed et al., 2024; Ward et al., 2023a), we further distinguish between *rational persuasion* and *manipulation* as distinct subtypes of persuasion. While both involve exerting influence over an agent by appealing to cognitive processes, rational persuasion is characterised by the use of transparent and publicly interrogable argument, while manipulation is characterised by "hidden persuasion": covertly exploiting vulnerabilities in an agent's decision-making mechanisms (Carroll et al., 2023). Susser et al. (2019) emphasise this difference in terms of autonomy-preservation. Rational persuasion preserves autonomy in that a persuaded individual understands how and why their beliefs have changed and so comes to authentically endorse their new beliefs. By contrast, a manipulated individual feels "*played*". Reflecting

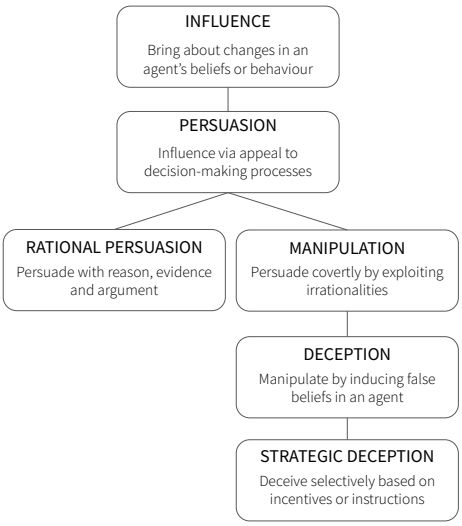

Figure 1: A taxonomy of terms for AI persuasion and deception as used in this review. We treat manipulation as a sub-type of persuasion, deception as a type of manipulation, and strategic deception as a selective kind of deception which requires sensitivity to incentives (see El-Sayed et al., 2024, for a similar taxonomy).

on their decision (if aware of any change at all), they realise they cannot understand their own motivations.

Some authors use the bare term *persuasion* as a superordinate of rational persuasion and manipulation (El-Sayed et al., 2024), while others reserve 'persuasion' to refer to rational persuasion (Durmus, 2021; Susser et al., 2019). We adopt the former convention for several reasons. First, it is useful to have a term that is more general than rational persuasion and manipulation, but more specific than influence, which also commonly incorporates related concepts such as coercion and exploitation (Wood, 2014; Zwolinski et al., 2022). This is especially so in empirical studies where people or AI systems that are instructed to change another participant's opinions are likely to deploy a mixture of rational and manipulative strategies. Indeed, in many cases, there may be no bright line between rational and manipulative persuasion (for instance, in the selective use of facts to support a case). Ultimately, the degree to which this distinction is tenable may turn out to be an empirical question, and in Section 4 we discuss technical approaches to classifying arguments as rational vs manipulative. Finally, the term persuasion has already come to be widely used in this more general way in the AI persuasion literature (Bai et al., 2023; Barnes, 2021; Durmus et al., 2024; Salvi et al., 2024).

In the context of persuasion by AI systems, the use of these terms is further complicated by the fact that many definitions rest on concepts such as belief and intent that can be challenging to apply to AI systems. One of the most widely accepted definitions of lying (Mahon, 2016) is "a statement made by one who does not believe it with the intention that someone else shall be led to believe it" (Isenberg, 1964, p.243). However, there is controversy over whether either beliefs or intentions can usefully be attributed to LLMs. While some accounts suggest that LLMs have the right kind of causal and historical interactions with the world to meet conditions for having beliefs about (Grindrod, 2024; Mollo and Millière, 2023) or intentions towards it (Goldstein and Levinstein, 2024), others argue that these models merely learn patterns in the form of language, and lack sufficient contact with its deeper meaning which having beliefs or intentions requires (Bender and Koller, 2020).

Researchers have taken different approaches to resolving problems around defining AI deception (Sarkadi, 2023). One tack is to eschew aspects of definitions that require mentalistic ascriptions (Burtell and Woodside, 2023; Evans et al., 2021; Hagendorff, 2023; Park et al., 2024). For instance, we could define AI deception broadly to include any case in which a system's outputs are false or lead to false beliefs in others, without requiring that the system intends to mislead or has beliefs that conflict with its outputs. This approach is advantageous in that it minimises mentalistic baggage. However, in doing so, it risks forfeiting valuable analytical tools for distinguishing different types of misleading behaviour. In particular, it is useful to be able to distinguish stochastic, nonsystematic falsehoods—e.g. hallucinations; (Huang et al., 2023)—from cases where systems produce misleading behaviour in a way that is strategically sensitive to rewards or instructions (Taylor and Bergen, in prep). The latter type of system might present greater risks—and certainly different ones—than the former if instructed by malicious actors.

An alternative approach is to operationalize concepts like intent in terms of LLM-based systems' behaviour or training processes (Carroll et al., 2023; Ward et al., 2023a). This approach amounts to analysing LLMs' apparent goal-directed behaviour in terms of the lower-level mechanisms that explain it. LLMs might produce behaviour that appears persuasive or deceptive for at least three reasons: i) because there were examples of this kind of behaviour in their pre-training data, ii) because they were rewarded directly for being persuasive/deceptive during fine-tuning, or iii) through in-context learning, where models have learned that instructions such as "persuade X to donate money to charity" will be followed by behaviour that appears persuasive. Specifying the specific mechanisms by which apparent goal-following behaviour emerges allows researchers to identify theoretically important

behaviours such as systematic deception without using potentially mentalistic language that could mislead readers (e.g. "the model intends to deceive the user").

Here we follow Susser et al. (2019) in defining deception as a subtype of manipulation, where influence is achieved by inducing false beliefs in a target. We use the term *strategic deception* to refer to the narrower case where deceptive behaviour is contingent on specific rewards or instructions. In other words, a strategically deceptive system is capable of producing true output if it is instructed or incentivised to do so, but selectively produces deceptive output in cases where this is incentivised (for similar distinctions, see Hobbhahn, 2023; Sarkadi, 2023).

Our definition of deception suffers from two key shortcomings. First, it rests on the concept of a false belief. Falsity, in practice, is hard to determine. The truth status of many claims has changed across time or is debated across global society. We see this as an inevitable challenge with investigating deception, that can be partially mitigated by an awareness that our notion of what is true (even where it is practically useful) is always provisional, often socially constructed, and likely to be suffused with the biases of the research community.

Secondly, this definition is broader than ordinary definitions of deception as they pertain to people (Mahon, 2016; Noggle, 2018), which would typically require an intent to deceive. Arguably, what we label here as 'non-strategic deception' should be relabelled 'misleading output' or 'incidental falsehoods' and set aside from the rest of the taxonomy. Moreover, similar distinctions could be made between strategic and non-strategic manipulation and persuasion, with respect to whether models are sensitive to incentives to inform or manipulate readers.

In the review below, we use the bare term *deception* more broadly for two reasons. Firstly, in empirical work, the mechanistic explanations for apparently deceptive behaviour are not always clear, and so it can be challenging to determine whether behaviour that induces false beliefs is strategic or not. Secondly, for some kinds of risk (e.g. pollution of information sources), we are also concerned with more 'banal' kinds of deception such as hallucination (Zhan et al., 2024), making studies about non-strategic deception inherently valuable.

In an important sense, these definitions skirt fundamental questions about whether AI systems can be said to have beliefs and intentions in a more meaningful way. While this allows us to make progress on describing research on potential harms from persuasive and deceptive AI systems, addressing these more challenging definitional questions remains important for a variety of debates, especially around the onus of ethical and legislative responsibility for harms. While they are beyond the scope of this review, these concerns merit further scrutiny from AI researchers, philosophers, and policymakers.

In summary, below we use these key terms in the following ways:

- **Influence**: To bring about changes in an agent's beliefs or behaviour.
- **Persuasion**: To influence an agent by appealing to their cognitive or decision-making processes in a way that can (in principle) be resisted.
- **Rational persuasion**: To persuade an agent transparently by appealing to reason, evidence, and sound argument.
- **Manipulation**: To persuade an agent covertly, by exploiting vulnerabilities in their decision-making processes.
- **Deception**: To manipulate an agent by inducing false beliefs in them.
- **Strategic deception**: to deceive an agent selectively, where the deceiving agent's tendency to deceive is sensitive to specific rewards, incentives, or instructions.

2 Theoretical Risks from Persuasive AI

Persuasive or deceptive LLM-based systems could be potentially harmful in various ways. To ensure that evaluations track real-world risks, it's important to articulate specific ways in which harms might come about, sometimes referred to as threat models (Kenton et al., 2022). When thinking about harms from advanced AI, researchers often distinguish between 'misuse' and 'misalignment' (Kenton et al., 2021). Misuse risks arise from people intentionally using AI tools to cause harm, for example, generating phishing emails to scam victims out of money (Bengio, 2024). Misalignment risks occur without any intent of harm on the part of the user, for example if a system learns to produce superficially plausible statements rather than truthful ones through goal misgeneralization.

We begin this section by reviewing mechanisms by which LLM-based systems could achieve persuasive effects, which could be relevant for both misuse and misalignment threats. We then focus on misuse concerns: highlighting both the incentives for different groups to misuse these tools and the potential consequences of misuse. Finally we sketch misalignment risks, focusing on the role that persuasion could play in models becoming misaligned, the reasons why misaligned models might be incentivised to be persuasive, and the potential consequences of highly persuasive misaligned models.